Roy Zheng

+1 (604)-729-6358 | royyi2006@gmail.com | linkedin.com/in/roy-zheng-06rz29 | github.com/R0yZh3ng

Education

University of British Columbia — Vancouver, BC
Bachelor of Science in Mathematics — *Sept 2024 – April 2028*

Experience

Freelance Software Developer — Calgary, AB
Full-Stack Web Developer & Consultant — Peter's Home Solution — *Sept 2026 – Present*

- Designed and shipped a production website for a local home-services business (React, TypeScript, FastAPI, SQLAlchemy/PostgreSQL) in a **single Docker image** deployed on **Google Cloud Run** with a custom domain and auto-provisioned HTTPS
- Built an owner-facing admin dashboard behind a secret, non-guessable URL with API-key auth, constant-time comparison, and a **login-attempt lockout**; added a Pillow-backed photo upload pipeline that re-encodes and **strips EXIF/GPS metadata** from every uploaded image before it's served
- Built a client-side **bilingual (English/Chinese) PDF** invoice generator (jsPDF) with exact integer-cent arithmetic and a server-issued, **concurrency-safe** auto-incrementing invoice number, keeping all customer billing data off the server by design

Projects

OSCAR: Wall-Projected Assistant | *Python, Flask, SQLite, JavaScript, Raspberry Pi, systemd* — Sept 2026 – Present

- Built a Raspberry Pi hub that projects a self-written habit tracker (Flask, SQLite, vanilla JS) onto a wall and switches pages on an RFID tap; every state change goes through one **named-action dispatch** function with an audit log, and **63 unit tests** run with no hardware via injected dependencies
- Traced slow taps to page load, not the display link, and replaced relaunch-per-tap with a kept-alive Chromium that **activates preloaded tabs** over its remote-debugging API; four services start together at boot as **systemd** units
- Implemented per-wall keystone correction: a **4-point homography** (DLT, Gaussian elimination) emitted as a CSS `matrix3d` on a wrapper page, re-warping every open tab live via **Server-Sent Events**, with a drag-the-corners calibration page

Fourier Analysis Engine for Financial Markets | *Python, NumPy, Pandas, SciPy* — Apr 2026 – Jun 2026

- Engineered a spectral signal processing pipeline applying **Welch's Power Spectral Density** over raw FFT to extract dominant market cycles from equity price data; reduced noise via windowed averaging across overlapping segments, improving frequency resolution for low-SNR time series
- Developed a walk-forward backtesting framework with sliding window validation across **220 historical trades**, achieving **49.1%** directional accuracy on TSLA, **48.2%** on SPY and **46.4%** on MAGS; interpreted near random performance as empirical evidence consistent with the Efficient Market Hypothesis

Hackathons

PictureTs | *React Native, TypeScript, AWS Bedrock, Skia* | Kickstart '25, **6th Place** — Nov 2025

- Built an AI-powered mobile camera assistant in a team of **4** that provides live camera composition feedback and auto-adjusts zoom, exposure, and flash by piping compressed viewfinder frames through **AWS Bedrock (Nova Lite)** and parsing structured JSON commands from prompt-engineered multimodal responses
- Engineered a **custom image processing pipeline** using React Native Skia to apply brightness, contrast, and saturation via color matrix transformations, after identifying that `expo-image-manipulator` lacked support for low-level pixel operations

LeGenesis | *React, React Three Fiber, FastAPI, SQLAlchemy, Supabase, MeshyAI* — Mar 2026

- Built an interactive sketch canvas (React, React Three Fiber) with **split-screen editing**, enabling users to draw concepts and instantly visualize AI-generated 3D models
- Developed a FastAPI backend integrating MeshyAI's text-to-image and image-to-3D pipeline, with a stateless **context-token system** for **fast iterative edits** without database round-trips
- Implemented an **agentic planning layer** using Fetch.ai (uAgents) to classify prompts and intelligently route generation through optimized pipelines (single-view vs. multi-view) based on desired speed/quality tradeoffs

Technical Skills

Programming: Python, C/C++, Java, TypeScript, JavaScript, SQL, React, React Native, FastAPI, Flask
Quantitative: Signal Processing, Spectral Analysis, Time Series Modelling, Backtesting, Market Microstructure
Machine Learning: PyTorch, scikit-learn, Autoencoder Architectures, Anomaly Detection, Gradient-based Optimization
Mathematical: Linear Algebra, Numerical Methods, Probability, Dynamical Systems
Systems & Tools: Linux, Git, Docker, AWS, Google Cloud Run, Raspberry Pi, systemd, NeoVim

UBC Science Co-op
science.coop@ubc.ca | 604-822-9677